Sentiment Analysis: Is Android better than iPhone?

CSC601: Data Analytics and Visualization

by

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Github Repository Link:

https://github.com/girishsakpal/49_Girish_Sakpal_DAV_Assignment_02

1. Introduction

In today's digital era, social media platforms like Twitter play an important role in expressing public opinion. Users frequently share their views on trending topics, including comparisons between popular smartphone platforms such as Android and iPhone. These opinions provide valuable insights into user preferences and market trends.

This project focuses on performing sentiment analysis on tweets related to the topic *"Is Android better than iPhone?"*. Sentiment analysis is a Natural Language Processing (NLP) technique used to classify text into different categories based on its emotional tone. In this project, tweets are classified into positive, negative, and neutral sentiments.

The main objective is to apply and compare different machine learning models, including Naive Bayes, Support Vector Machine (SVM), and Logistic Regression, to evaluate their performance using metrics such as accuracy, precision, recall, and F1 score.

The project also emphasizes data analytics and visualization, helping to better understand the dataset, preprocessing steps, and model performance.

2. Dataset Description

The dataset used in this project consists of approximately 100 tweets collected based on the topic *Android vs iPhone*. These tweets reflect user opinions, experiences, and comparisons between the two smartphone platforms.

Each record in the dataset contains the following fields:

- **Text**: The actual content of the tweet, representing user opinions in natural language.
- **Sentiment Label**: A predefined category indicating the sentiment of the tweet. The labels include:
  - **Positive**: expressing favorable opinions
  - **Negative**: expressing criticism or dissatisfaction
  - **Neutral**: expressing neither strong positive nor negative sentiment

Key Characteristics of the Dataset

- The dataset contains around 100 tweets, making it suitable for a small-scale sentiment analysis task.
- The distribution of sentiment classes is approximately balanced, ensuring that no single class dominates the dataset.
- The data was cleaned and pre-processed by removing missing values and duplicate entries to improve quality and consistency.
- The dataset was divided into two subsets:
  - Training Set (80%): used to train the machine learning models
  - Testing Set (20%): used to evaluate model performance

Unlike many sentiment analysis tasks, this dataset already included pre-labeled sentiment values, eliminating the need for manual annotation. This allowed the focus of the project to remain on preprocessing, model building, and performance evaluation.

3. Methodology

The project was carried out in multiple stages, starting from data preparation to model evaluation and comparison. Each step was designed to ensure accurate sentiment classification and meaningful analysis.

3.1 Data Collection & Cleaning

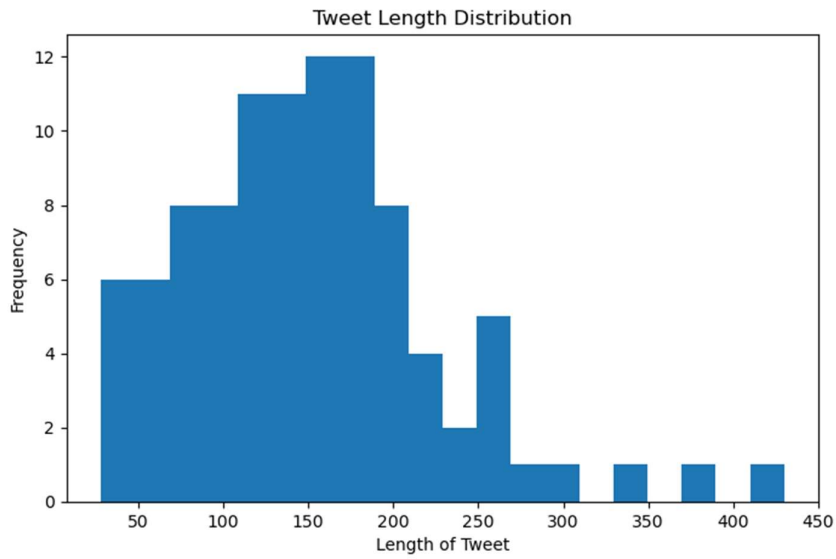
The dataset was loaded from a CSV file containing tweets related to the Android vs iPhone comparison. Initial data cleaning was performed to ensure data quality and consistency.

The following steps were applied:

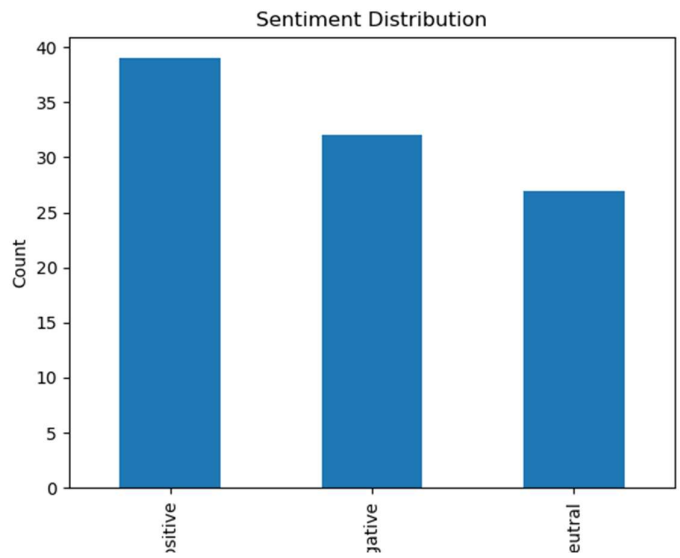
- Removed missing values to avoid incomplete data during processing
- Eliminated duplicate tweets to prevent bias in model training
- Selected only relevant columns: text and label
- Renamed the column label to sentiment for clarity and consistency
- Limited the dataset to 100 tweets to meet assignment requirements

Visualization:

- **Tweet Length Distribution**: To analyze variation in tweet sizes



- **Sentiment Distribution**: To observe class balance



3.2 Text Preprocessing

To prepare the textual data for machine learning models, several Natural Language Processing (NLP) techniques were applied:

- Converted all text to lowercase for uniformity
- Removed unwanted elements such as URLs, User mentions (@username), Hashtags (#), Numbers and Punctuation marks

- Removed stopwords (common words like “the”, “is”, etc.) that do not contribute to sentiment
- Applied lemmatization to reduce words to their base form (e.g., “running” → “run”)

Feature Engineering:

- Created a new column clean_text containing processed text
- Compared original vs cleaned text length to analyze preprocessing impact

3.3 Data Splitting

The dataset was divided into training and testing sets:

- **80% Training Data**: Used to train the models
- **20% Testing Data**: Used to evaluate model performance

A stratified sampling technique was used to ensure that the proportion of sentiment classes (positive, negative, neutral) remained consistent in both training and testing datasets.

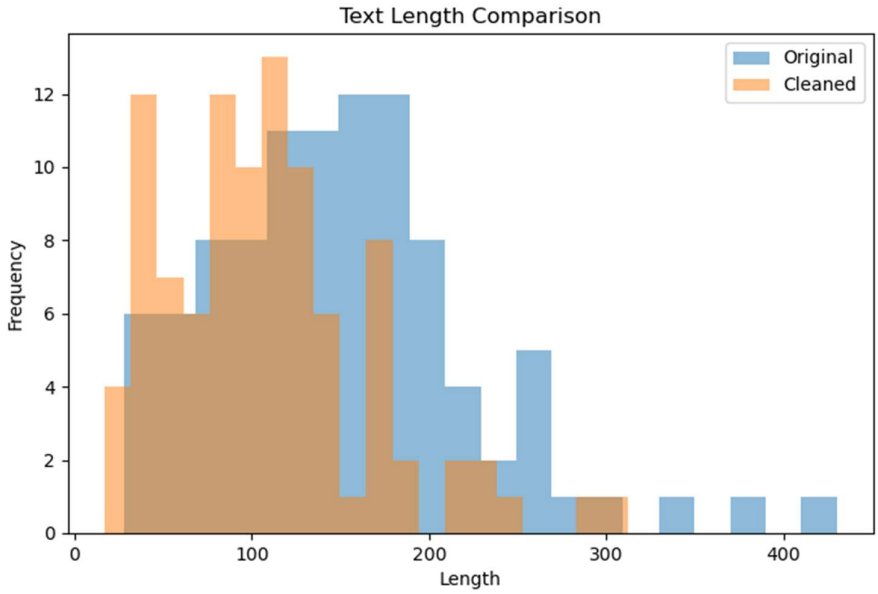
3.4 Feature Extraction

Text data was converted into numerical form using TF-IDF (Term Frequency - Inverse Document Frequency):

- Assigns weights to words based on their importance
- Reduces the influence of common words and highlights meaningful terms

The following configurations were used:

- Unigrams and bigrams to capture both single words and word pairs
- Maximum feature limit to optimize performance and reduce dimensionality



3.5 Model Implementation

Three machine learning models were implemented and trained on the processed data:

- **Naive Bayes:** A probabilistic classifier that assumes feature independence and works efficiently for text classification tasks.
- **Support Vector Machine (SVM):** A model that separates classes by finding the optimal hyperplane, especially effective in high-dimensional data.
- **Logistic Regression:** A linear model that predicts class probabilities and provides balanced and reliable classification performance.

3.6 Evaluation Metrics

The models were evaluated using multiple performance metrics:

- **Accuracy**: Overall correctness of predictions
- **Precision**: Correctness of positive predictions
- **Recall**: Ability to identify actual positives
- **F1 Score**: Harmonic mean of precision and recall

- **Confusion Matrix**: Visual representation of prediction errors
- **Classification Report**: Detailed metric breakdown for each class

3.7 Model Comparison

To determine the best-performing model, a comprehensive comparison was conducted using:

- Accuracy
- F1 Score
- Precision and Recall

Additionally, a **weighted final score** was calculated to provide a more balanced evaluation:

$$FinalScore = 0.4(F1) + 0.3(Precision) + 0.3(Recall)$$

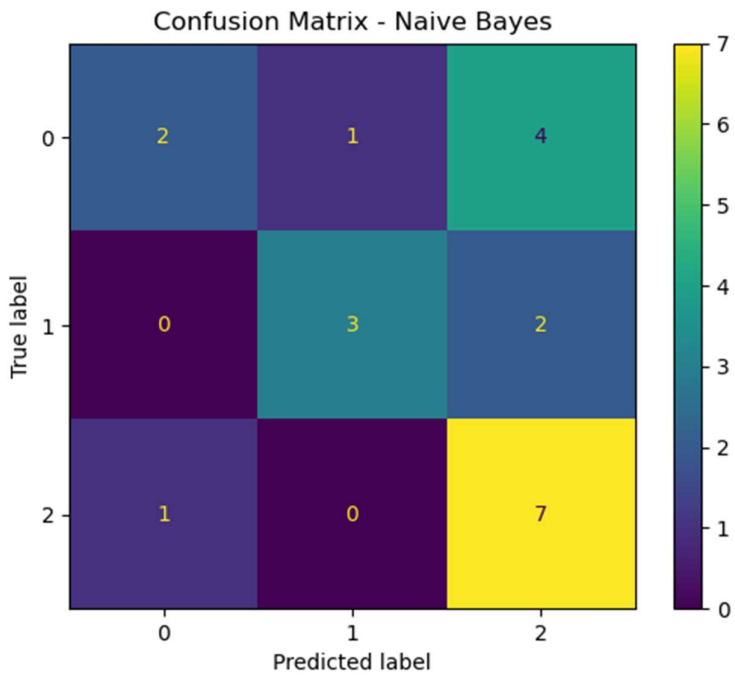
This approach ensures that the model selection is not solely based on accuracy but also considers the balance between precision and recall.

4. Results

The performance of the three machine learning models, namely, Naive Bayes, Support Vector Machine (SVM), and Logistic Regression, was evaluated using accuracy, precision, recall, and F1 score. These metrics provide a comprehensive understanding of how well each model performs on the test dataset.

4.1 Naive Bayes

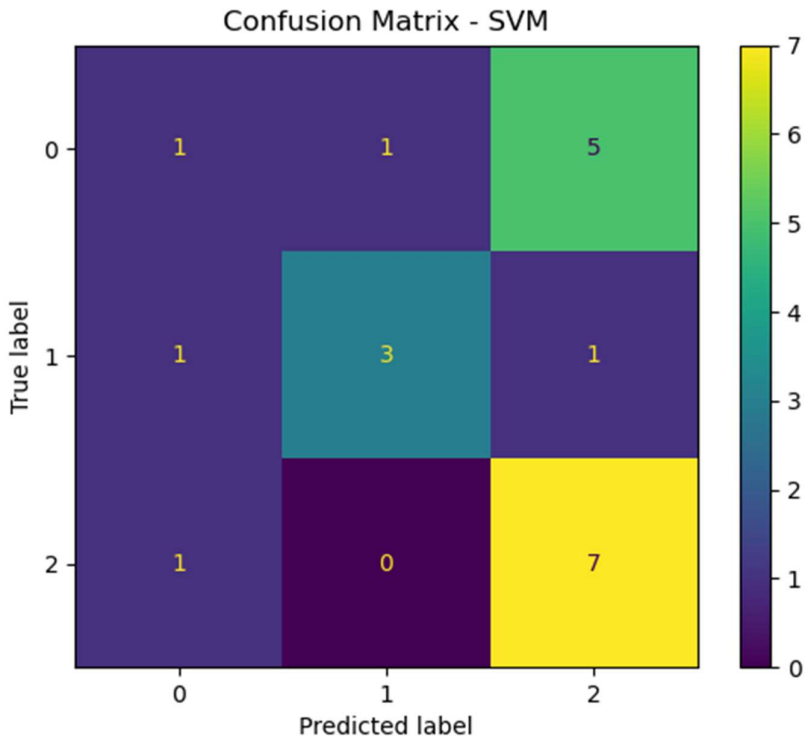
- **Accuracy:** 0.60
- **Precision:** 0.636
- **Recall:** 0.60
- **F1 Score:** 0.573



The Naive Bayes model achieved a moderate accuracy of 60%, indicating that it correctly classified a majority of the test samples. It showed relatively balanced performance across precision and recall, resulting in a higher F1 score compared to other models. This suggests that Naive Bayes performs consistently across different sentiment classes and is effective for basic text classification tasks.

4.2 Support Vector Machine (SVM)

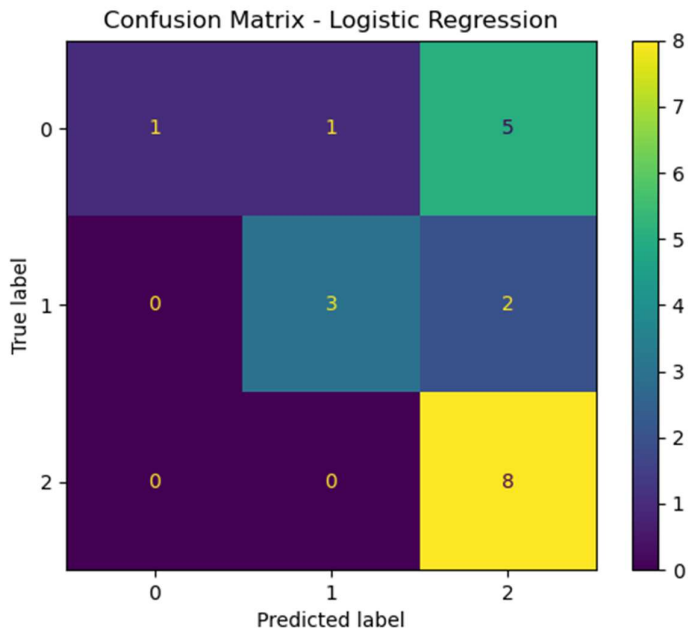
- **Accuracy:** 0.55
- **Precision:** 0.519
- **Recall:** 0.55
- **F1 Score:** 0.503



The SVM model achieved the lowest performance among the three models, with an accuracy of 55%. Both precision and F1 score are comparatively lower, indicating that the model struggled to correctly classify some sentiment categories. Although SVM is generally powerful for high-dimensional data, its performance in this case may be affected by the small dataset size.

4.3 Logistic Regression

- **Accuracy:** 0.60
- **Precision:** 0.750
- **Recall:** 0.60
- **F1 Score:** 0.532

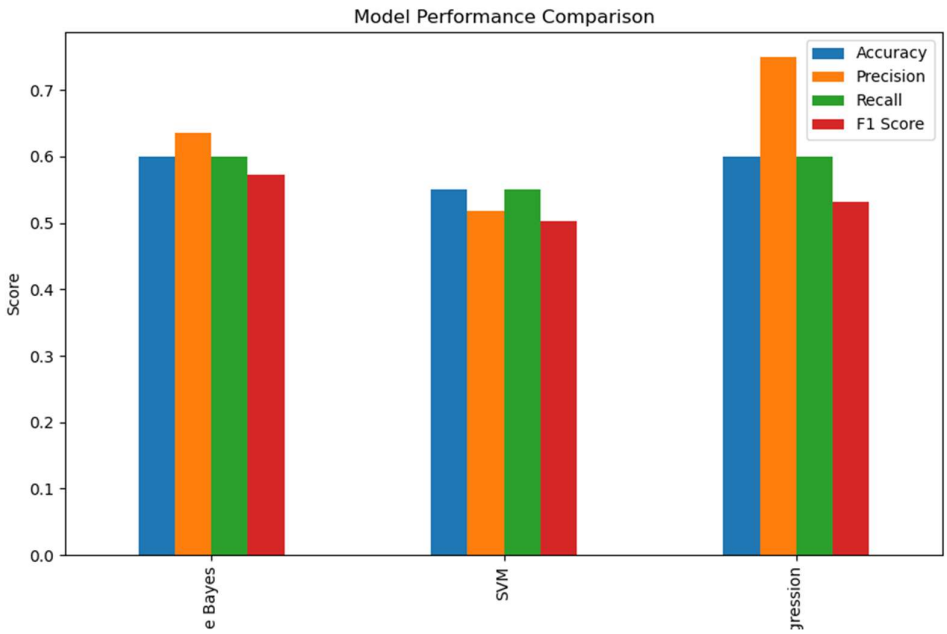


Logistic Regression achieved the same accuracy as Naive Bayes (60%) but showed significantly higher precision. This indicates that when the model predicts a particular sentiment, it is more likely to be correct. Although its F1 score is slightly lower than Naive Bayes, the higher precision reflects better reliability in predictions.

4.4 Model Comparison

Model	Accuracy	Precision	Recall	F1 Score	Final Score
Naive Bayes	0.60	0.636	0.60	0.573	0.6000
SVM	0.55	0.519	0.55	0.503	0.5219
Logistic Regression	0.60	0.750	0.60	0.532	0.6178

A detailed comparison shows that while both Naive Bayes and Logistic Regression achieved equal accuracy, their performance differs when considering other metrics. Naive Bayes has a slightly higher F1 score, indicating better balance between precision and recall. However, Logistic Regression has significantly higher precision and the highest overall weighted score.



To make a more reliable decision, a Final Score was calculated using a weighted combination of F1 score, precision, and recall. This approach ensures a balanced evaluation rather than relying solely on accuracy.

5. Conclusion

This project successfully implemented sentiment analysis on tweets comparing Android and iPhone using multiple machine learning models. The tweets were classified into positive, negative, and neutral sentiments, and the models were evaluated using accuracy, precision, recall, and F1 score.

Both Naive Bayes and Logistic Regression achieved the same accuracy of 0.60. However, Logistic Regression showed higher precision and a better overall balance between evaluation metrics, resulting in the highest final weighted score. In comparison, SVM performed slightly worse due to lower precision and F1 score.

This project highlights that accuracy alone is not sufficient for model selection. A deeper evaluation using multiple metrics is necessary to determine the most reliable model.

Based on the overall comparison, **Logistic Regression achieved the highest Final Score (0.6178)** and is selected as the best model. It provides the most balanced and consistent performance, making it the most suitable choice for this sentiment analysis task.